Chapter

Biochar-Enhanced Diets in Naeemi Sheep: A Strategy for Enteric Methane Mitigation and Growth Optimization

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Abstract

This study aims to evaluate biochar's potential to reduce methane emissions that contribute to global warming and to examine its effects on enteric fermentation. Recent estimates show that agricultural methane emissions in the EU-15 are roughly 10.2 million tons annually, making it the largest source of such emissions. About two-thirds of these originate from enteric fermentation, while one-third comes from livestock manure. In this research, methane emissions from ruminants were measured in two groups of Naeemi ewes. The treatment group received a diet with added biochar alongside a standard diet, while the control group was given only a standard diet. On day 0, methane emissions of 1,070 ppm were recorded. The addition of biochar to the diet significantly reduced CH_4 emissions by 70.79–74.24% ($P < 0.05$). Additionally, animals in the treated group showed statistically significant improvements in body weight ($P < 0.05$), measuring 64.19 kg, along with notable increases in body length, heart girth, and height at the withers, which were 80.49 cm, 82.95 cm, and 100.49 cm, respectively, compared to the control groups. Biochar appears to have the potential to enhance nutrient utilization, improve feed efficiency, and reduce methane emissions in animals. As a feed supplement, biochar may positively influence digestion and improve the ruminant's capacity for nutrient absorption.

Keywords: sheep, methanogenesis, global warming, biochar, greenhouse gases

1. Introduction

Global food security and sustainable agriculture are urgent issues amid rising demand and climate change. While expanding crop and livestock production is vital for meeting nutritional needs [1], the environmental impacts, especially from the livestock sector, must be addressed. Livestock accounts for approximately 14.5% of total human-caused greenhouse gas (GHG) emissions (~7.0 – 7.1 Gt CO_2 equivalent/year), with methane (CH_4) and nitrous oxide (N_2O) being the main pollutants [2].

Methane from enteric fermentation alone constitutes nearly 39–40% of livestock GHG emissions, with the remainder originating from manure, feed production, and land-use practices [3, 4]. Since methane has 28–36 times higher global warming potential than CO_2 over 100 years, reducing emissions in livestock systems is crucial [5].

Several dietary strategies are under study, including the use of biochar, a carbon-rich material produced by pyrolyzing organic waste at 250–600°C in low-oxygen environments [6]. Meta-analyses and recent trials indicate that biochar can improve nutrient utilization, feed efficiency, and gut health, while potentially reducing methane emissions in ruminants [7, 8].

A 2024 study on specific biochars for cattle confirmed positive effects on rumen fermentation and microbial populations, although the impact on CH_4 emissions varied depending on the type and dose of biochar [9]. Similarly, a 2022 study published by Jindo et al. found that reducing methane emissions through the addition of biochar in compost made from poultry and cattle decreased emissions by 3.0 times for poultry compost but by 3.7 to 4.6 times for cattle compost [10]. A broader review in Animal Feed Additives (2023) reported improvements in digestion, weight gain, and methane reduction across various livestock species.

Mechanistic studies suggest that biochar's high surface area and cation-exchange capacity facilitate the adsorption of hydrogen ions (H^+) produced during enteric fermentation. The shift in microbial metabolism from methane generation to ammonium production results in ammonium adsorption onto biochar surfaces, thereby decreasing CH_4 emissions [11, **12**]. The effectiveness of biochar depends significantly on its properties, including particle size, porosity, and surface chemistry [2, 8].

This study investigates the effects of sterile date-palm biochar added to sheep diets, hypothesizing that it will suppress enteric methane emissions by diverting fermentation processes toward ammonium production; enhance rumen fermentation efficiency, nutrient utilization, and animal health; and improve fecal consistency and manure quality. By achieving environmental and productivity goals, this research contributes to climate-smart livestock production and sustainable diets.

We propose that incorporating biochar into livestock feed could potentially reduce methane emissions while also promoting animal welfare.

2. Materials and Methods

2.1 Animals

The field study was conducted at the Kuwait Institute for Scientific Research (KISR) facilities. A total of 28 adult ewes were randomly divided into two equal groups, with 14 ewes per group. Each ewe was housed in an individual enclosure and received two equal portions of feed per day, specifically at designated times in the morning and evening, totaling 0.8 kg of feed per day for each individual. There were no significant differences between the two groups in any respect; however, one group was fed a standard diet alone, while the other group was fed a standard diet supplemented with biochar.

2.2 Preparation of standard diet

Preparation of the standard diets, which consist of 70% concentrates and 30% roughage, is essential.

2.3 Production of biochar

Date palm fronds were obtained from the Kuwait Institute for Scientific Research and then reduced to the appropriate sizes for preparing biochar. The biochar was prepared using a thermochemical process called pyrolysis, which occurred at a high temperature of 550°C in an oxygen-free environment. Subsequently, the biochar underwent grinding and sieving procedures using a mill to obtain uniformly sized particles that were less than 2 mm in diameter [12].

2.4 Preparation of new ration

Standard pellet diet was prepared by passing it through a 2 mm sieve to ensure uniformity in the size of the grains, facilitating effective blending. Subsequently, biochar was added to the prepared ration at a level of 1%.

2.5 Methane gas emissions

The study was conducted within the Kuwait Institute for Scientific Research facilities. Methane emissions quantification was carried out using respiration chambers and methodological and conceptual frameworks that were previously developed during the 2011 and 2012 greenhouse gas experiments [7, 13].

Fourteen female Naeemi sheep, aged between 10 and 12 months, were kept in respiration chambers. Following the outlined protocol in [7], methane emissions were measured.

Before collection, each sheep underwent a 21-day acclimatization period, during which time it was offered a basal diet. After acclimatization, methane emissions were recorded over nine weeks.

The sheep received food twice a day in equal quantities at staggered times: 5:00 a.m. (hour 0) and 11:00 a.m. (hour 6). The subjects were randomly assigned to two groups of seven individuals each to ensure equal allocation.

2.6 Statistical analysis

The results obtained from the field experiment were analyzed using GraphPad Prism 10 software. The treatment values were compared against the control group values. Differences in variance were evaluated using one-way ANOVA, where statistical significance was set at the $P < 0.05$ level.

3. Results

3.1 Standard diet prepared

The standard diet's overall composition consisted of 70% concentrates and 30% roughage. Details regarding the formulation of the standard diet are provided in **Figure 1**.

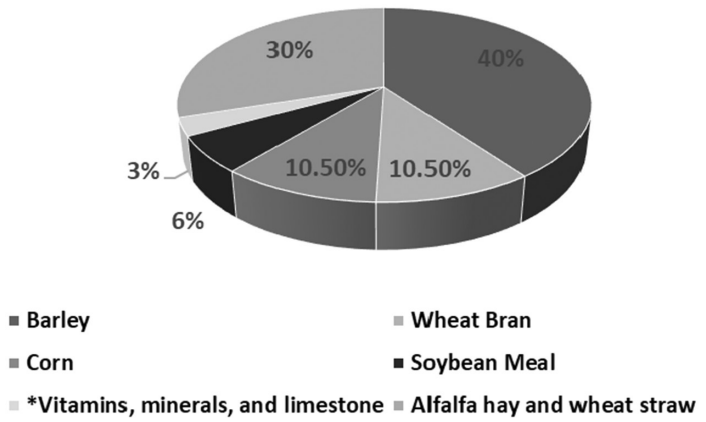


Figure 1.
*The experimental basal ration, tailored to the specific nutritional needs of Naeemi sheep and expressed on a dry matter basis, consists of the following ingredients: barley (40%), wheat bran (10.5%), corn (10.5%), soybean meal (6%), and a premix containing vitamins, minerals, and limestone (3%), as well as a mixture of alfalfa hay and wheat straw (30%). *the premix composition includes phosphorus (5%), calcium (18%), sodium (5%), potassium (5%), manganese dioxide (500 mg/kg), cobalt sulfate (100 mg/kg), zinc oxide (2000 mg/kg), calcium iodate (125 mg/kg), sodium selenate (10 mg/kg), vitamin a (400,000 IU/kg), vitamin D_3 (100,000 IU/kg), and alpha-tocopherol (400 iu/kg) (source: developed by the authors).*

3.2 Patterns of growth, development, and weight gain

Results showed that adding biochar to the basal diet significantly ($P < 0.003$) enhanced body live weight, along with height at withers (HAW), body length, and heart girth (HG) or chest circumference in Naeemi sheep compared with the control.

Mean body live weight for sheep on the basal diet with biochar addition was 64.19 kg, compared with 59.81 kg for the control (**Figure 2**).

Significant progress was also noticed in growth parameters within the biochar application treatment plot; these sheep measured 80.49 cm in body length, 82.95 cm in heart girth, and 100.39 cm in height at the withers.

On the other hand, the corresponding measurements for the control were 77.23 cm, 79.99 cm, and 96.31 cm (**Figure 2**).

3.3 Body condition score (BCS)

Animals following a basic feeding protocol with biochar supplementation showed a significantly higher average body condition score (BCS) of 3.5, compared to a control BCS of 2.76.

Numerous studies have irrefutably demonstrated that adding biochar to the diets of sheep has a positive effect on fat deposition. Biochar, a carbon-rich solid material made from organic matter, has been found to be effective in increasing nutrient utilization efficiency along with feeding efficiency in ruminant species.

When incorporated into the feeding plan for these animals, biochar can modify digestive processes, ultimately improving nutrient absorption efficiency, especially for fats, which, in turn, promotes increased fat deposition within the animals (**Figure 2**).

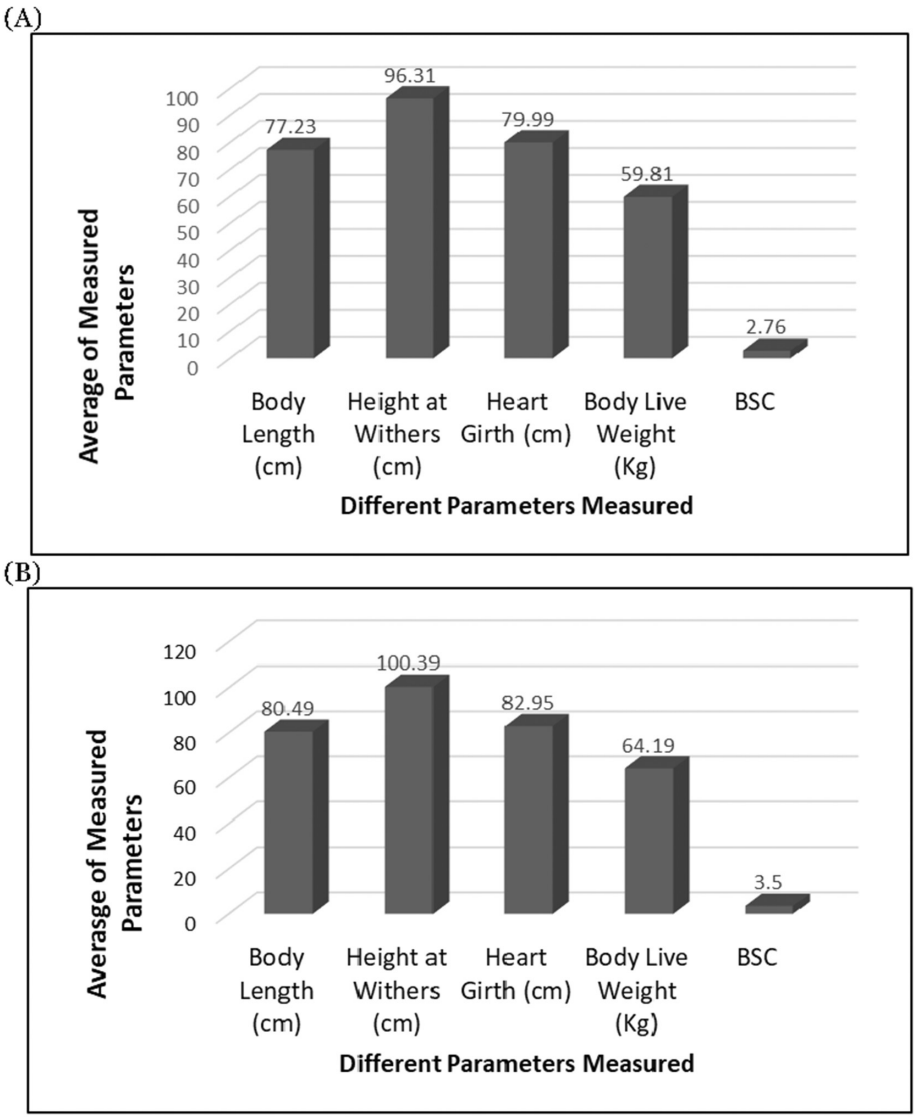


Figure 2.
Average body length, height at withers, heart girth, body live weight, and body condition score (BCS) of Naeemi sheep fed (a) a basal ration (control) and (b) a basal ration supplemented with biochar during the feeding trial. Values are expressed as means (source: developed by the authors).

3.4 Methane emission control

Table 1 shows that on Day-0, immediately before biochar was introduced into the diets of the subjects, the baseline methane emission rate observed across all participants under the two experimental conditions was 1,070 ppm.

The experimental group, whose diets were supplemented with biochar, had a significantly lower methane production rate ($P < 0.05$), recording declines of 71.52% and 74.24% in weeks 1 and 9, respectively, compared to the control group.

Duration in weeks	Standard diet (Control)	Standard diet + Biochar	% of CH4 reduction
Day-0	1,070 ± 0.21	1070 ± 0.21	-
1	1,072 ± 0.02	625 ± 0.03	71.52
2	1,074 ± 0.01	626 ± 0.04	71.57
3	1,076 ± 0.14	630 ± 0.98	70.79
4	1,072 ± 0.22	620 ± 0.19	72.90
5	1,074 ± 0.31	622 ± 0.23	72.67
6	1,077 ± 0.10	624 ± 0.87	72.59
7	1,078 ± 0.23	625 ± 0.67	72.48
8	1,077 ± 0.02	630 ± 0.32	70.95
9	1,096 ± 0.03	629 ± 0.04	74.24

Table 1.
Methane (CH_4) gas production (ppm) in Naeemi sheep fed a basal ration (control) or a basal ration supplemented with biochar over a 9 week feeding period. Values are presented as mean ± SEM, with percentage CH_4 reduction calculated relative to the control group (source: developed by the authors).

During the experimental period, the mean SMC in the control group did not change at all, remaining at 1,076.6 ppm, with recorded fluctuations between 1,070 and 1,096 ppm, and no statistically significant differences were noted over time ($P > 0.05$).

In contrast, the experimental group exhibited a significant and sustained decline in methane emissions after the introduction of biochar. Emission rates consistently decreased from a baseline of 1070 ppm on Day 0 to an average of 629 ppm by Week 9. The decline was found to be statistically significant ($P < 0.05$) compared to both the control group and the baseline measurements, ranging from 70.79% to 74.24%.

3.5 Total feed consumption

(**Figure 3**) presents the average feed conversion ratio (FCR) for all animal groups used in the study (kg/2 s). FCR is a key determinant of feeding efficiency, particularly in assessing study (kg/2 week). FCR is a key determinant of feeding efficiency, especially with the ability of livestock to transform feed into body mass, such as meat.

FCR for the control group was observed to be 16.87 ± 0.92. The biochar-supplemented group, however, showed a significantly higher FCR value of 18.99 ± 0.68.

The purpose of the present study was to design a novel dietary regimen for sheep to reduce methane emissions caused by ruminant fermentation. Observational data showed that Naeemi sheep treated with a supplemented diet of biochar exhibited significantly enhanced body live weight and several growth parameters, including body length, height at withers, heart girth, and body condition score, compared with the control group. These observations align with findings reported in earlier studies [8].

A noteworthy feature of the present study's outcomes is the evidence of enhanced feed efficiency resulting from supplementation with biochar.

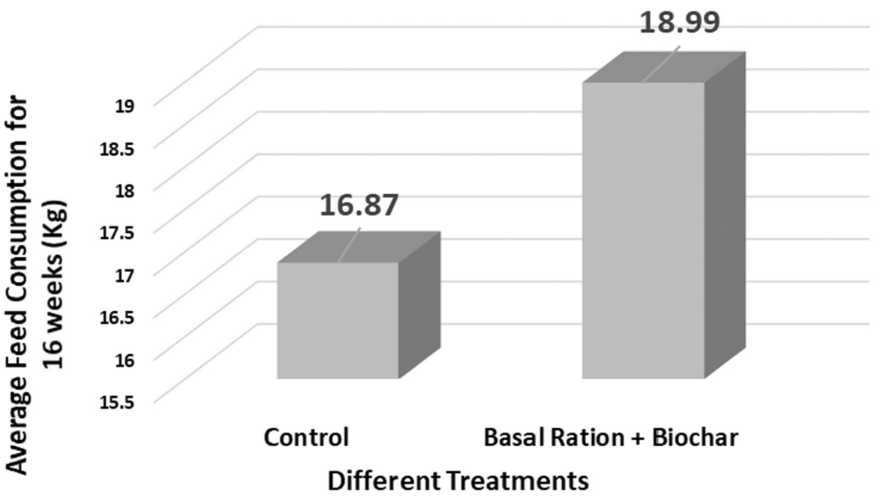


Figure 3.
The average feed consumption ratio (kg) of Naeemi sheep across a 16-week feeding period was determined when they were fed a basal diet (control) or a basal diet supplemented with biochar. The biochar supplementation of the feed increased feed intake from 16.87 to 18.99 kg/2 weeks in comparison to the control feeding regimen (source: developed by the authors).

Biochar improves the absorptive capacity and digestibility of essential nutrients, resulting in increased assimilation and greater availability of these nutrients. Moreover, biochar acts as a binder, thereby reducing excretory losses and enhancing nutrient utilization. Existing literature substantiates a connection between biochar application and increased weight gain, as well as overall growth in livestock [14].

4. Discussion

The major objective of this study was to develop a modified feeding regime for sheep to substantially reduce methane emissions during ruminant digestion. The experiment involved two different groups of Naeemi sheep: one group was fed a biochar-enriched diet, and the other group was fed a standard diet.

The results showed that the sheep from the biochar-enriched feeding regime exhibited significantly greater body mass and growth parameters, including body length, height at the withers, and heart girth, compared to the control group. These findings support previous observations reported by [15]. Such improvements are largely attributed to biochar's ability to enhance the breakdown and assimilation of nutrients, thereby optimizing feeding efficiency.

In addition, biochar helps to decrease nutrient leaching, which optimizes feed utilization, leading to increased weight gain and growth in livestock, as reported in previous research [15]. Moreover, biochar has been associated with improved gastrointestinal health [16, 17], as it enhances the proliferation of beneficial gut microorganisms, stimulates immunity, and reduces the occurrence of various diseases.

Importantly, biochar can adsorb harmful contaminants, such as toxins, heavy metals, and mycotoxins in the gastrointestinal tract before they can enter the bloodstream, thus leading to better health in livestock.

It is essential to recognize that excessive use of biochar can restrict root penetration and result in variations in nutrient availability, as reported by some previous studies [18, 19].

4.1 The effects of incorporating biochar in livestock feeds.

Earlier studies have shown that including bamboo biochar at a dose of 1 g/kg of live weight produces significant results [6, 20, 21]. Goats fed diets supplemented with biochar experienced an increase in crude protein intake and improved nitrogen absorption, leading to lower nitrogen excretion through urine and feces.

Additionally, the goats in the study showed a notable rise in their average daily weight gain, increasing by 53 grams from the 44 grams recorded in the control group. To gain further insights into this phenomenon, the study used a controlled feeding regime primarily involving tannin-rich *Acacia mangium* leaves to determine if digestive efficiency was improved through the combined effects of tannins and biochar, thereby increasing the availability of protein and dry matter for better nutritional uptake.

In a parallel study involving 12 groups of goats, researchers compared growth performance between scenarios with a 1% biochar addition to a standard mixture of *Bauhinia acuminata* leaves – known for their high tannin content – and scenarios without biochar. The addition of biochar in this setup resulted in better nutrient absorption, leading to a 27% increase in daily weight gain over the 100-day feeding period. However, the results presented here show some inconsistencies regarding the reported improvements.

Furthermore, in another specific study on goats fed diets with 1.5% activated coconut biochar as well as with 3%, no variation in feeding rates or microbial community compositions was observed when compared with control animals [21]. Such contradictory results show that the effectiveness of biochar supplementation is highly dependent on numerous factors, such as the type of livestock involved, the nutrient composition of the feeds used, the choice of biochar type, the dosages applied, and the experimental conditions employed.

The impact of biochar addition on the feed conversion ratio (FCR) shows variation with respect to species. Some research findings suggest a reduction in the FCR due to biochar addition. For instance, in broiler chickens, biochar addition increased feed conversion efficiency while simultaneously inducing a reduction in overall food intake, thereby producing a better FCR ratio [8]. Similarly, the addition of biochar to breeding rabbit diets resulted in enhanced feeding efficiency, accompanied by a decline in food intake [14]. Conversely, other studies reported negligible or no significant effects.

Studies involving monogastric animals often show different results. In 2025, a study on mature pigs treated with commercial feed supplemented with 2% biochar found no statistically significant differences in average daily feed intake (ADFI), average daily weight gain (ADWG), and feed conversion ratio (G:F); however, there was a notable improvement in crude fiber digestion [22].

A more detailed study conducted in 2025 also yielded varied results, indicating that the effect of biochar on feed efficiency and growth performance is not always positive for monogastric animals [23]. In contrast, results for ruminant animals have generally been more promising.

In a highly controlled crossover trial with lactating dairy cows in late 2024, feeding 1% biochar had no significant effect on feed intake, milk yield, quality, or enteric methane emissions. Conversely, a meta-review in 2025 reported that adding biochar to various livestock species, such as fattening bulls, pigs, broilers, and sheep, consistently improved feed conversion efficiency, digestive function, and body weight gain, while also reducing enteric methane emissions (by up to 30–40% in cattle) [24, 25].

These contrasting findings highlight a key point: the effectiveness of biochar supplementation largely depends on animal species, baseline diet and health, biochar source and dosage, and experimental conditions. While responses in monogastric animals vary, outcomes in ruminants seem more favorable but are still context-dependent. [25]

4.2 Effect of adding biochar to sheep diets on body conditions

The animals that received biochar-supplemented diets had a significantly higher mean body condition score (BCS) value of 3.5, as opposed to a BCS value of 2.76 among the control animals. These findings align with those reported in [26], which solidify the health-improving effects of biochar on the deposition of adipose tissue in ruminant hosts. Biochar, defined as a substance derived from biomass that is similar to charcoal, has proven effective in improving nutrient utilization efficiency and enhancing feed conversion ratios in feedlot populations of livestock.

In sheep nutrition, biochar supplementation has been shown to influence digestive mechanisms, thereby enhancing nutrient absorption, particularly lipids. This increased nutrient uptake is thought to contribute to the enhanced fat deposition observed in biochar-fed animals. The current study supports these findings.

For example, a previous study conducted on Awassi lambs receiving a 1% biochar dosage for 90 days showed improvements in carcass fat percentage and intramuscular fat deposition when compared to control groups, without showing any adverse effects on measures of meat quality [17, 27].

Another study carried out on Barki sheep showed that adding bamboo-derived biochar to their diet significantly improved fat deposition in adipose tissue and seemed to maximize energy use efficiency [16, 26]. Though the precise mechanisms of action are unknown, more recent studies have noted several contributing factors: enhanced nutrient assimilation, favorable changes in rumen populations of microbes (fostering bacteria with lipid-synthesizing capability), and improved barrier integrity of the intestine.

Overall, across multiple experiments with lambs and mature sheep fed varying biochar levels (typically 0.5–2% of diet dry matter), a consistent trend emerges: increased fat deposition and carcass lipid content, supporting biochar's potential role as a feed additive to improve body condition and productivity in small ruminants.

4.3 Minimization of greenhouse gas emissions

The information contained in **Table 1** shows a significant reduction in methane gas emissions after adding biochar to livestock feeding diets. A baseline measurement of methane emissions was initially recorded at 1070 ppm in week 0.

From week 1 onwards, methane emissions were reduced in all treatment groups (**Table 1**), with reductions ranging from 70.79% to 74.24% when compared to the entire field experiment and the control plot. These results are consistent with previous studies carried out by [28], which illustrate that biochar is an effective measure for reducing methane emissions from livestock because it is thought to suppress methanogenic activity by altering gut microbial communities.

Findings of the current study concur with recent research indicating the potential of biochar inclusion to substantially reduce enteric methane emissions from livestock. For example, an extensive review concluded that adding biochar to animal diets improved digestion and microbial action, simultaneously achieving considerable reductions in methane production of between 30% and 40% in cattle, with significant reductions observed in goats and poultry as well.

Furthermore, another experimental study undertaken in 2025 showed that biochar improved nutrient use, augmented feeding efficiency, and substantially reduced emissions of methane among diverse livestock tested [28]. However, variability cannot be overstated, as illustrated by a Swiss crossover study involving lactating cows given 1% biochar, which found no significant effects on methane outputs, milk yield, digestibility, and performance generally.

These inconclusive outcomes indicate that the effects of biochar inclusion on mitigating methane emissions are moderated by several variables, including the quality and type of biochar, the dosing rate used, dietary composition, animal species, and the environmental conditions under which they are kept.

Aside from its environmental importance, the use of biochar in lamb feeding has been linked to increased feed digestibility. Although this factor has been relatively less emphasized in recent studies, several studies have confirmed this benefit. A recent review once again linked biochar with improved feed digestion and nutrient absorption [28].

In agreement, a 2023 trial showed that the addition of corncob-based biochar led to a significant rise in dry matter intake, crude protein intake, and dry matter digestibility compared to control groups. Together, these studies highlight the twin merits of using biochar in feeding small ruminants: better environmental sustainability and increased nutritional effectiveness [29].

5. Conclusions

This study clarifies that the integration of sterile date-palm biochar into the feed of Naeemi sheep provides widespread benefits related to ecological sustainability and animal productivity. Specifically, the incorporation of 1% biochar into feed led to a significant reduction in enteric methane emissions, decreasing in the range of 70.79% to 74.24% across a nine-week feeding experiment, thus validating its role as a tool to curb greenhouse gas emissions from ruminant livestock.

Besides this specific environmental advantage, biochar-supplemented sheep demonstrated increased body weight, superior growth parameters, better body condition scores, and an increased feed conversion ratio in comparison to control groups. The results can be attributed to the high surface area and inherent cation-exchange capacity of biochar, which allow enhanced absorption of nutrients, increased efficiency of rumen fermentation, and favorable bacterial population substitutions leading to methanogenesis attenuation.

The discovery aligns with previous research examining the role of biochar in enhancing nutrient retention, increasing back fat deposition, and improving feed efficiency in ruminant species. Even though the results show promising implications, variability across previous research indicates the manifestation of varied factors, including types of biochar, biochar particle sizes, rates of use, types of diets, and the species-specific nature of the digestive system. More studies must be conducted to define optimal rates of use, desirable levels of integration, and to investigate wider implications for animal welfare, manure quality, and general agronomic productivity within the realm of integrated farming environments.

The outcome of the study supports the use of biochar as a multi-purpose feed supplement that ensures climate-resilient livestock production, improved animal performance, and, consequently, the creation of stronger and more resilient agricultural systems.

Conflict of interest

The authors declare no conflict of interest.

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